Status	Finished
Started	Monday, 26 October 2026, 6:58 PM
Completed	Monday, 26 October 2026, 7:02 PM
Duration	4 mins 22 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Searching
- c. Complexity
- d. Merging

The correct answer is:
Complexity

Question **2**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It makes the data unsorted
- c. It slows down the process
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **3**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It has no effect on the analysis process
c. It allows for quicker identification of duplicates
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort is difficult to implement
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Dividing the input into smaller subproblems
b. Sorting the entire list sequentially
c. Combining sorted sublists
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Bubbling up the smallest element
- c. Sorting the entire list in one pass
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **8**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Merging
- c. Searching
- d. Rearranging

The correct answer is: Sorting

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Bubble Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Selection sort
- c. Quick sort
- d. Merge sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting makes it harder to search for items
- b. Sorting decreases the efficiency of selection operations
- c. Sorting helps in finding duplicates quickly
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. arrange()
c. order()
d. sort()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection
b. Extraction
c. Insertion
d. Distribution

The correct answer is:
Insertion

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. The parts are ignored
c. Each part is searched for a specific element
d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **15**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Backtracking
- c. Dynamic programming
- d. Greedy method

The correct answer is:
Divide and conquer